

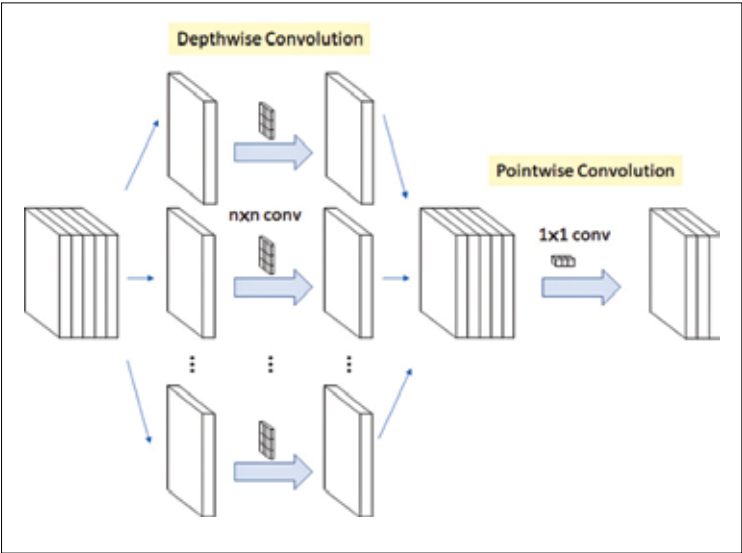


Figure : Depth wise and pint wise convolution

4. XCEPTION: Xception was proposed by François Chollet, the creator of the Keras library. Xception stands for Extreme version of Inception. With a modified depth wise separable convolution, it performs even better than Inception-v3 [2] for both JFT and ImageNet ILSVRC datasets. It was developed in 2017. The original depth wise separable convolution of the inception model is the depthwise convolution followed by a pointwise convolution in Xception model. They are described below:
1. Depth wise convolution is the channel wise $n \times n$ spatial convolution. For example, if we have five channels, then we will have five $n \times n$ spatial convolutions.

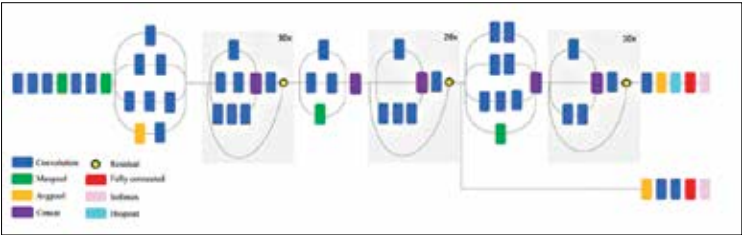


Figure : Inception Resnet V2 model